Seaborn

September 26, 2025

Seaborn

It is summarized that if Matplotlib "tries to make easy things easy and hard things possible", Seaborn tries to make a well-defined set of hard things easy too."

[1]:
```
pip install seaborn
```

```
Requirement already satisfied: seaborn in c:\users\omkar\anaconda3\lib\site-
packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in
c:\users\omkar\anaconda3\lib\site-packages (from seaborn) (2.1.3)
Requirement already satisfied: pandas>=1.2 in c:\users\omkar\anaconda3\lib\site-
packages (from seaborn) (2.2.3)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in
c:\users\omkar\anaconda3\lib\site-packages (from seaborn) (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in
c:\users\omkar\anaconda3\lib\site-packages (from
matplotlib!=3.6.1,>=3.4->seaborn) (1.3.1)
Requirement already satisfied: cycler>=0.10 in
c:\users\omkar\anaconda3\lib\site-packages (from
matplotlib!=3.6.1,>=3.4->seaborn) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in
c:\users\omkar\anaconda3\lib\site-packages (from
matplotlib!=3.6.1,>=3.4->seaborn) (4.55.3)
Requirement already satisfied: kiwisolver>=1.3.1 in
c:\users\omkar\anaconda3\lib\site-packages (from
matplotlib!=3.6.1,>=3.4->seaborn) (1.4.8)
Requirement already satisfied: packaging>=20.0 in
c:\users\omkar\anaconda3\lib\site-packages (from
matplotlib!=3.6.1,>=3.4->seaborn) (24.2)
Requirement already satisfied: pillow>=8 in c:\users\omkar\anaconda3\lib\site-
packages (from matplotlib!=3.6.1,>=3.4->seaborn) (11.1.0)
Requirement already satisfied: pyparsing>=2.3.1 in
c:\users\omkar\anaconda3\lib\site-packages (from
matplotlib!=3.6.1,>=3.4->seaborn) (3.2.0)
Requirement already satisfied: python-dateutil>=2.7 in
c:\users\omkar\anaconda3\lib\site-packages (from
matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
```

```
c:\users\omkar\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in
c:\users\omkar\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\omkar\anaconda3\lib\site-
packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

[2]:
```python
# Seaborn for plotting and styling
import seaborn as sns
df = sns.load_dataset('iris')  # Seaborn library having the predefined datasets
print(df.head())
```

```
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
```

[3]:
```python
len(df)
```

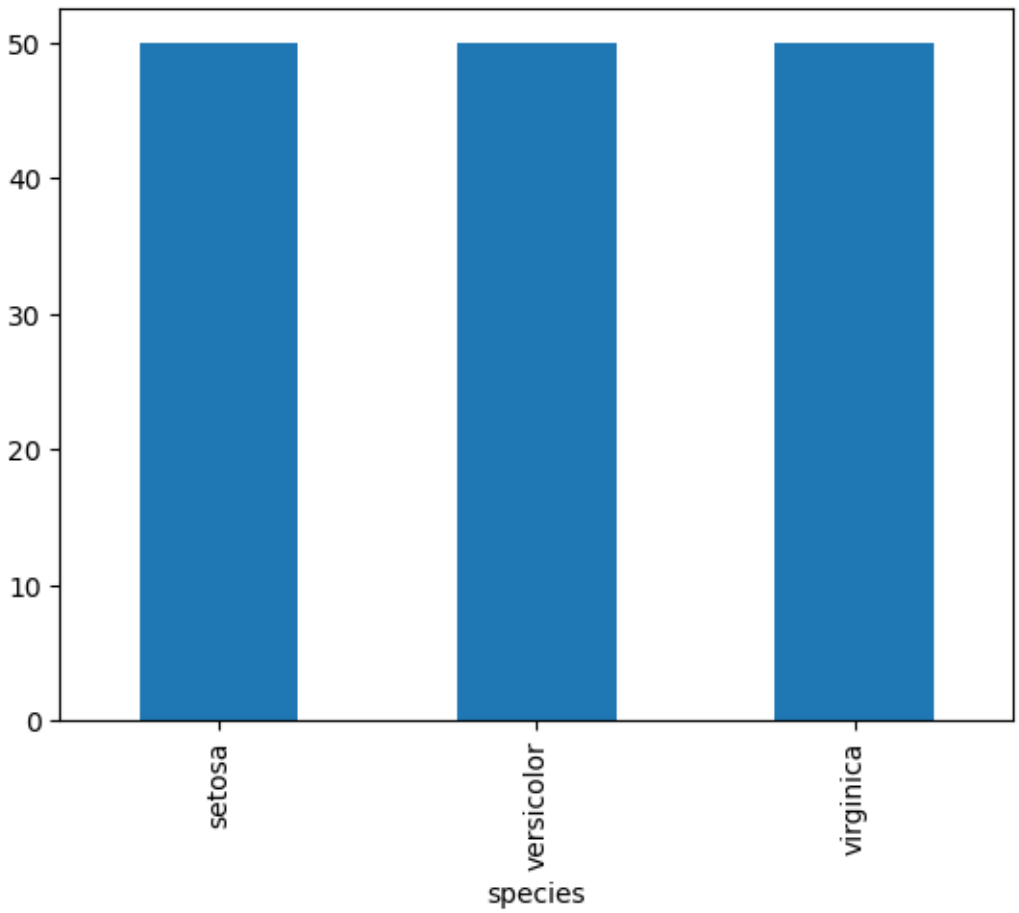
[3]:
```
150
```

[6]:
```python
df.species.value_counts()
```

[6]:
```
species
setosa        50
versicolor    50
virginica     50
Name: count, dtype: int64
```

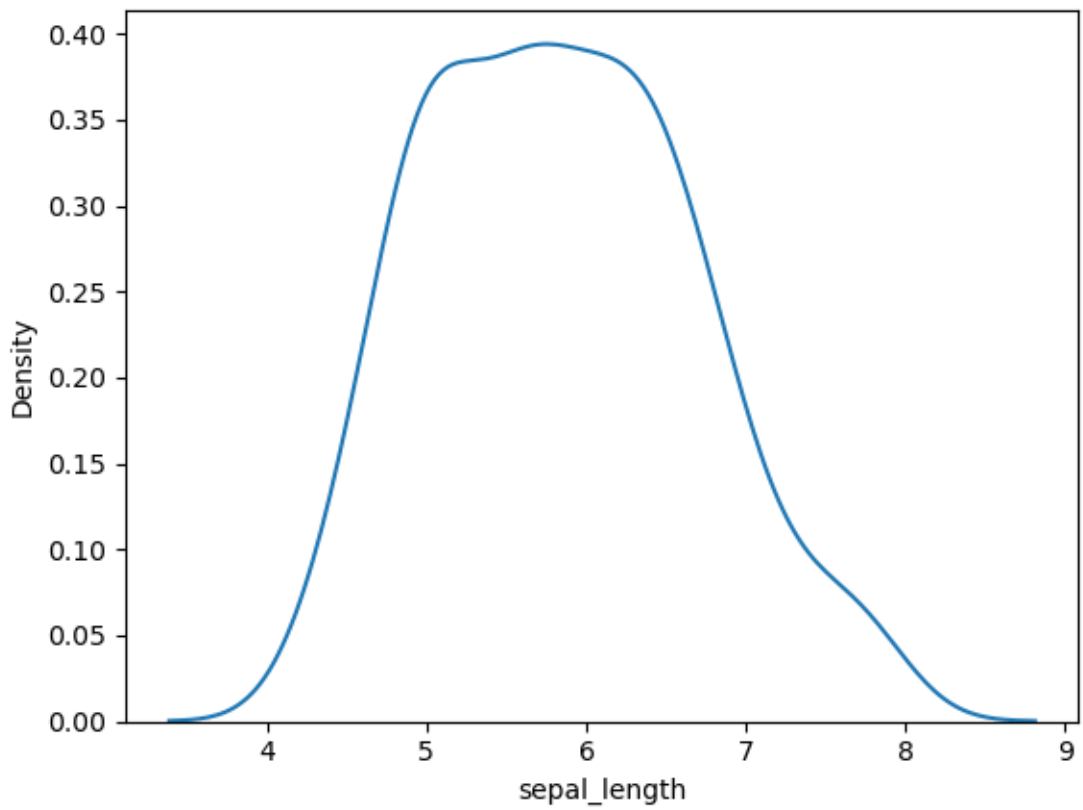
[7]:
```python
df.species.value_counts().plot(kind='bar')
```

[7]:
```
<Axes: xlabel='species'>
```



[8]:
```
sns.kdeplot(df['sepal_length'])
```

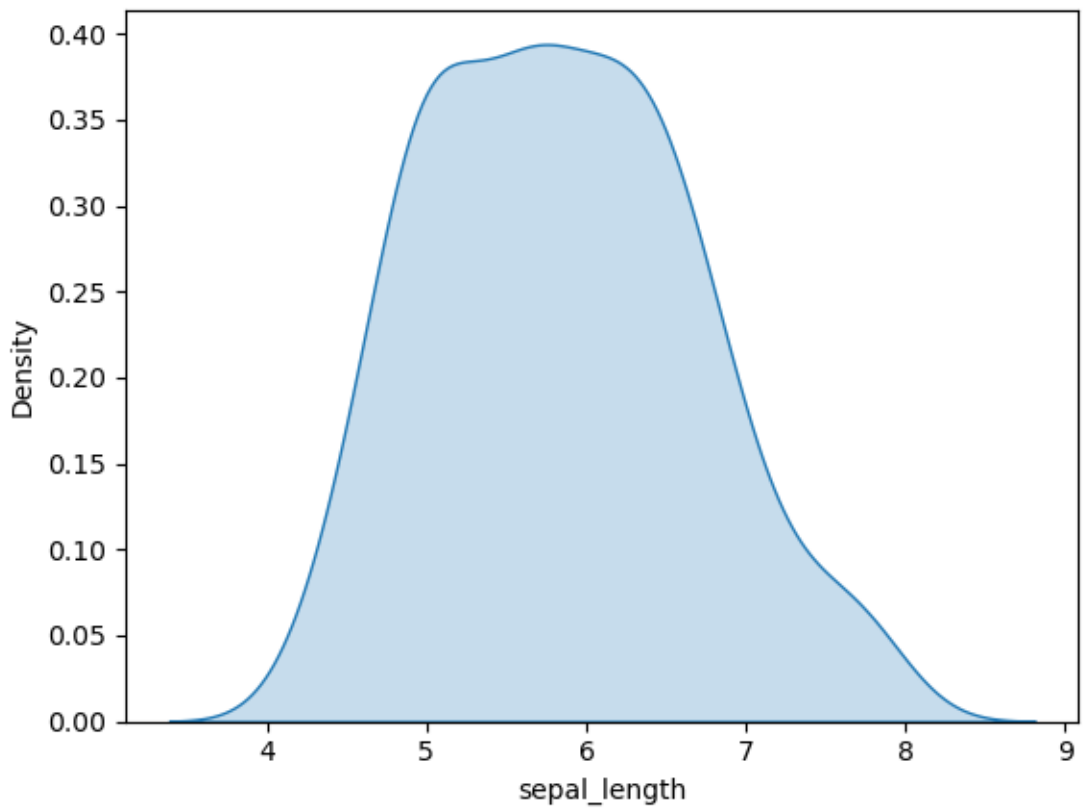
[8]: <Axes: xlabel='sepal_length', ylabel='Density'>



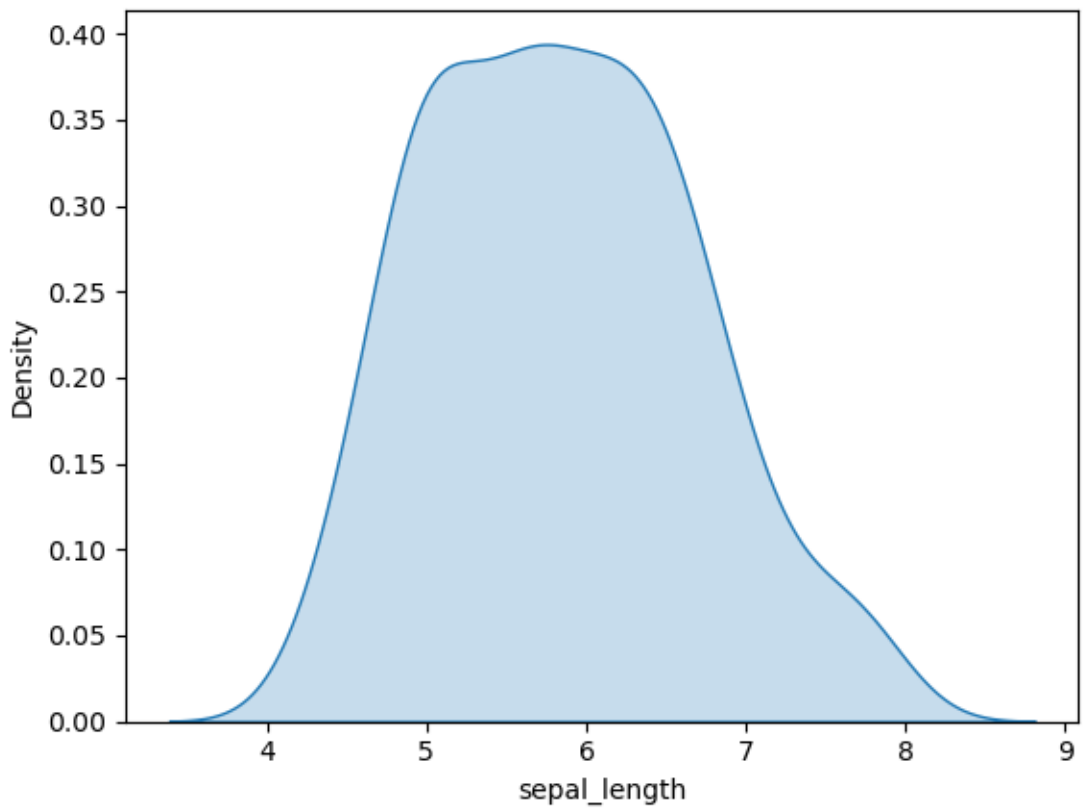
[10]:
```
sns.kdeplot(df['sepal_length'], shade=True)
```

```
C:\Users\OMKAR\AppData\Local\Temp\ipykernel_10392\3648197333.py:1:
FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

  sns.kdeplot(df['sepal_length'], shade=True)
```

[10]:
```
<Axes: xlabel='sepal_length', ylabel='Density'>
```



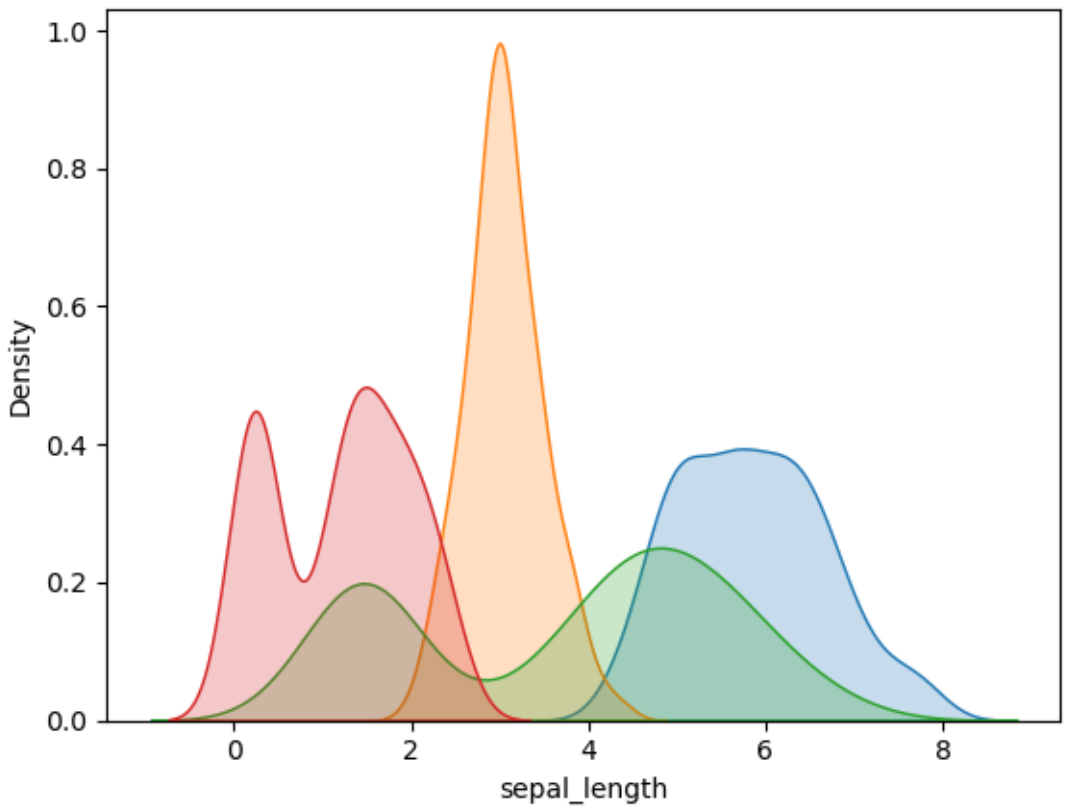
[13]:
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.kdeplot(df['sepal_length'], fill=True)
plt.show()
```



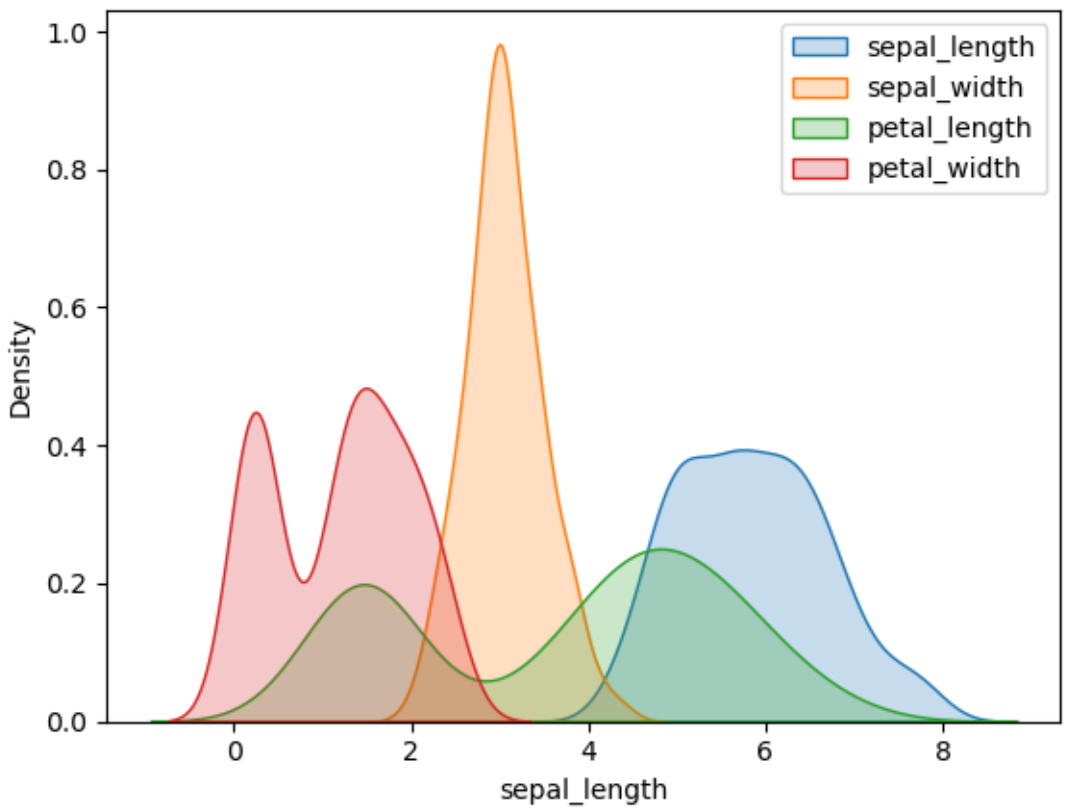
[14]:
```
# KDE: kernel density estimation
import seaborn as sns
import matplotlib.pyplot as plt
for col in ['sepal_length', 'sepal_width', 'petal_length', 'petal_width']:
    sns.kdeplot(df[col], shade=True)
```

```
C:\Users\OMKAR\AppData\Local\Temp\ipykernel_10392\667085740.py:5: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

  sns.kdeplot(df[col], shade=True)
C:\Users\OMKAR\AppData\Local\Temp\ipykernel_10392\667085740.py:5: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

  sns.kdeplot(df[col], shade=True)
C:\Users\OMKAR\AppData\Local\Temp\ipykernel_10392\667085740.py:5: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
```

```
This will become an error in seaborn v0.14.0; please update your code.

  sns.kdeplot(df[col], shade=True)
C:\Users\OMKAR\AppData\Local\Temp\ipykernel_10392\667085740.py:5: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

  sns.kdeplot(df[col], shade=True)
```



[15]:
```python
import seaborn as sns
import matplotlib.pyplot as plt

for col in ['sepal_length', 'sepal_width', 'petal_length', 'petal_width']:
    sns.kdeplot(df[col], fill=True, label=col)

plt.legend()
plt.show()
```

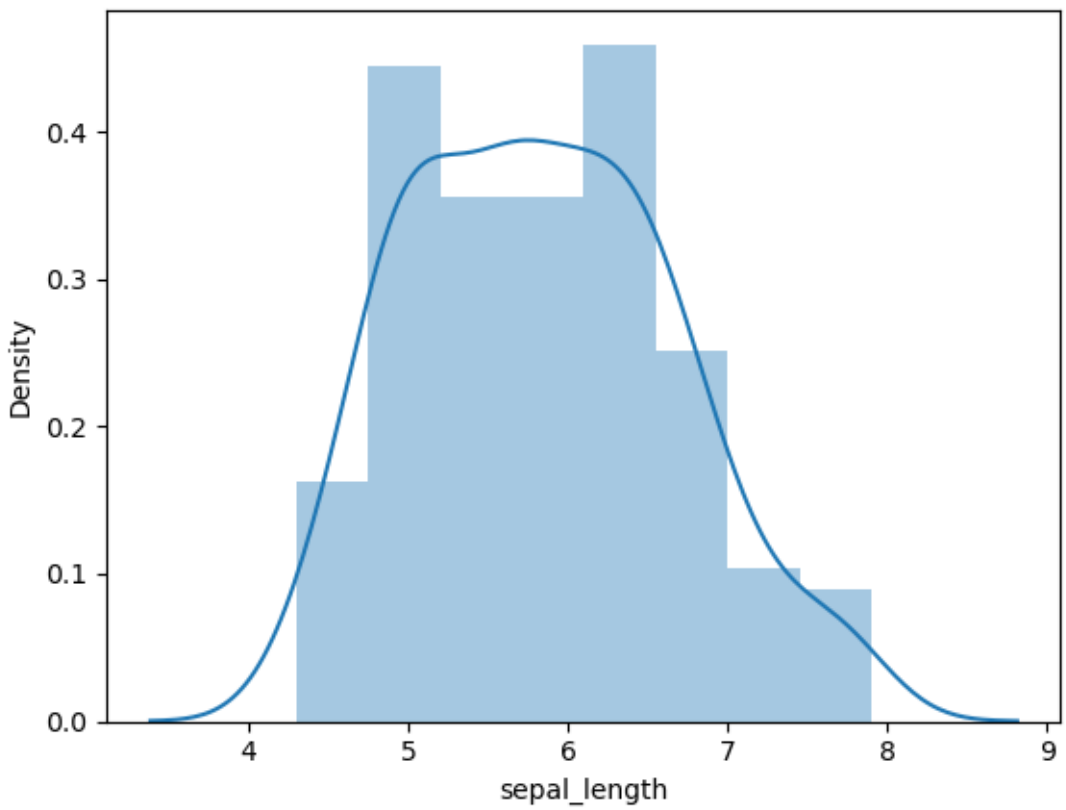


Histograms + KDE = Distplot

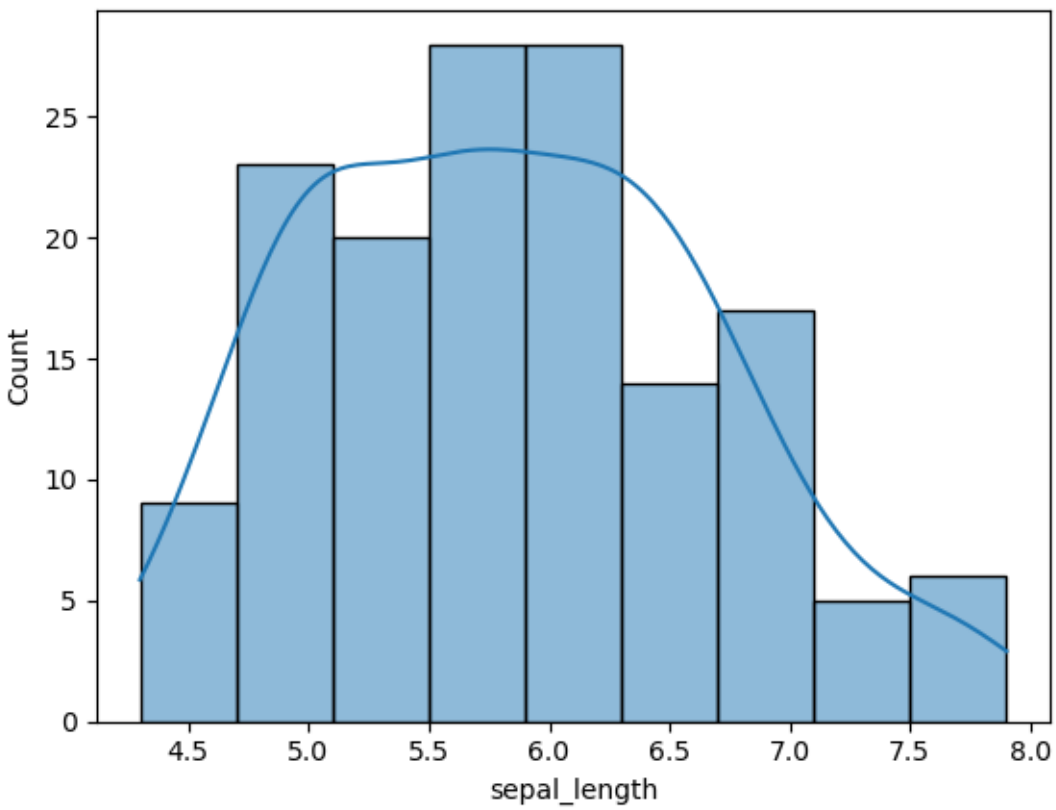
[16]:
```
sns.distplot(df['sepal_length'])
```

```
C:\Users\OMKAR\AppData\Local\Temp\ipykernel_10392\537145680.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df['sepal_length'])
```

[16]:
```
<Axes: xlabel='sepal_length', ylabel='Density'>
```

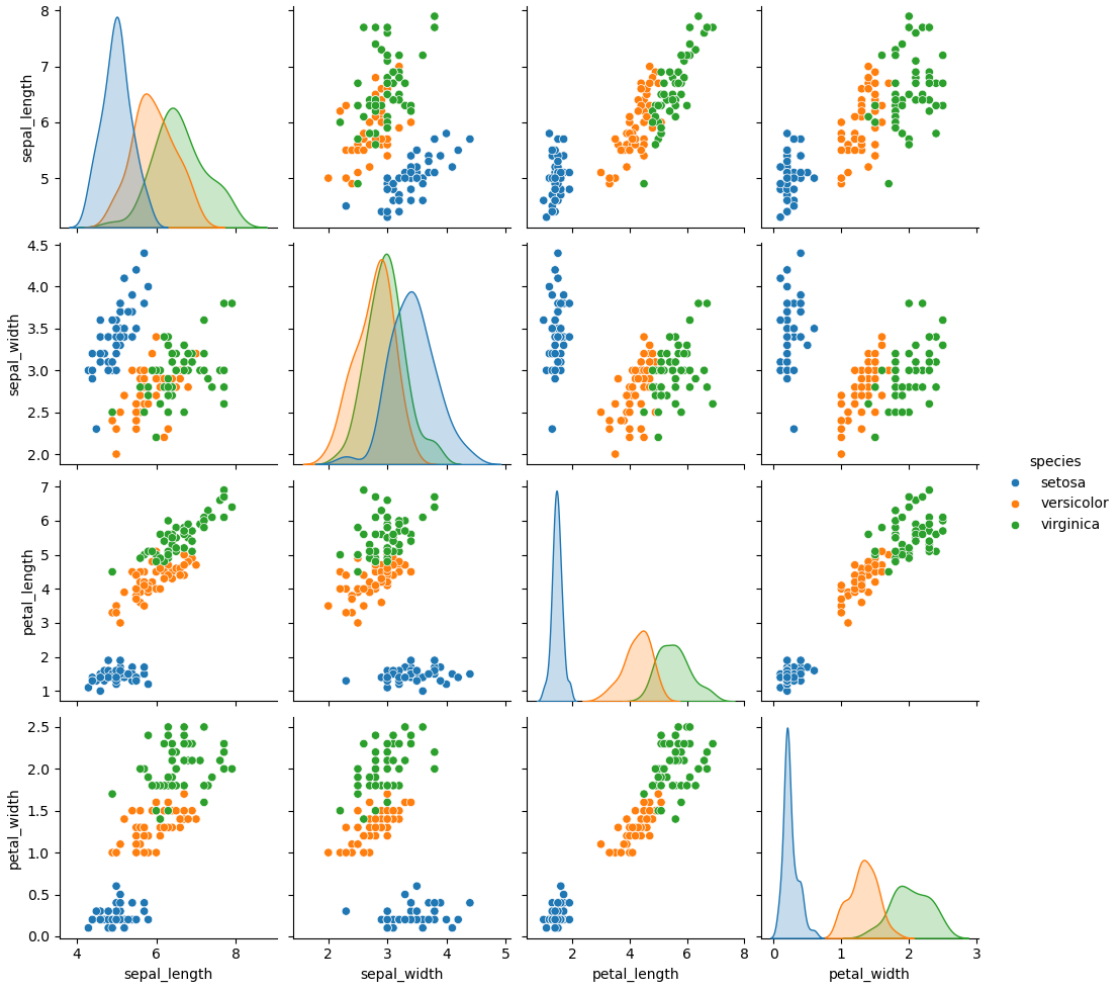


[17]:
```
import seaborn as sns
import matplotlib.pyplot as plt

# Replacement for distplot
sns.histplot(df['sepal_length'], kde=True)
plt.show()
```



[18]:
```
sns.pairplot(df, hue='species', size=2.5);
```

```
C:\Users\OMKAR\anaconda3\Lib\site-packages\seaborn\axisgrid.py:2100:
UserWarning: The `size` parameter has been renamed to `height`; please update
your code.
  warnings.warn(msg, UserWarning)
```



[19]:
```
Data = {'Year': [1920,1930,1940,1950,1960,1970,1980,1990,2000, 2010, 2020],
'Exchange Rate': [65, 69, 71, 64, 62, 59, 72, 71, 75, 78, 81]}
```

[20]:
```
Data
```

[20]:
```
{'Year': [1920, 1930, 1940, 1950, 1960, 1970, 1980, 1990, 2000, 2010, 2020],
 'Exchange Rate': [65, 69, 71, 64, 62, 59, 72, 71, 75, 78, 81]}
```

[21]:
```
type(Data)
```

[21]:
```
dict
```

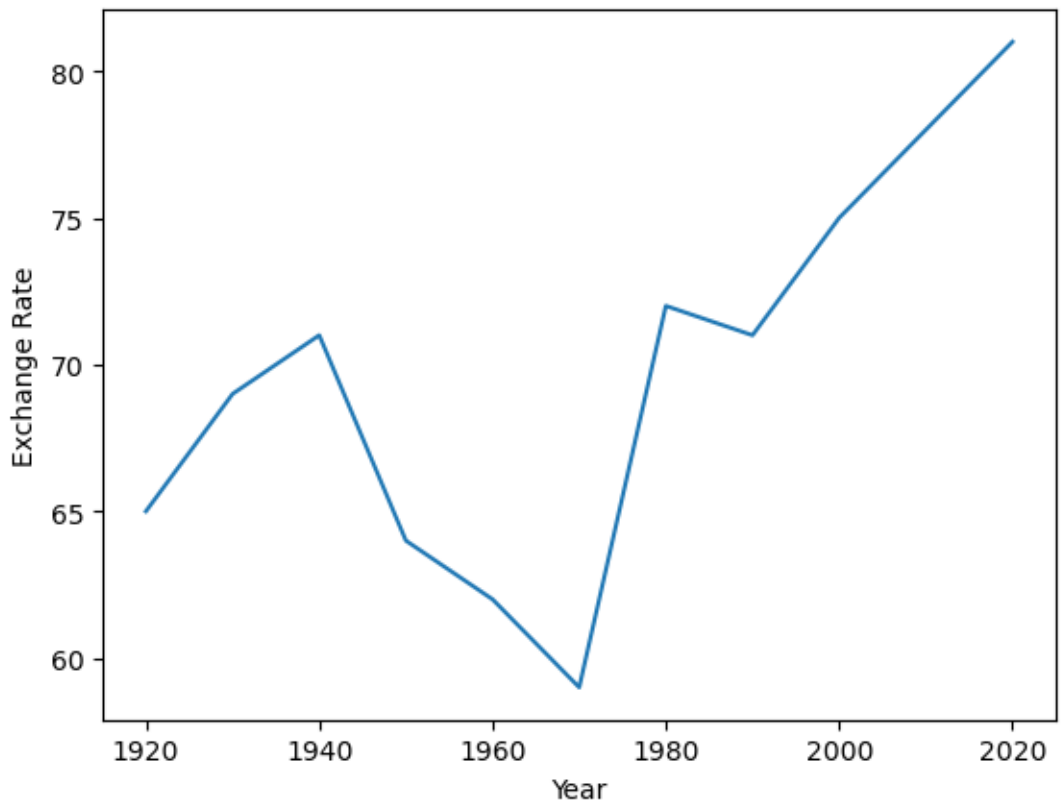
[23]:
```
import pandas as pd

df = pd.DataFrame(Data)
df.head(5)
```

[23]:

	Year	Exchange Rate
0	1920	65
1	1930	69
2	1940	71
3	1950	64
4	1960	62

[26]:
```
sns.lineplot(x = df["Year"], y = df["Exchange Rate"])
```

[26]: <Axes: xlabel='Year', ylabel='Exchange Rate'>



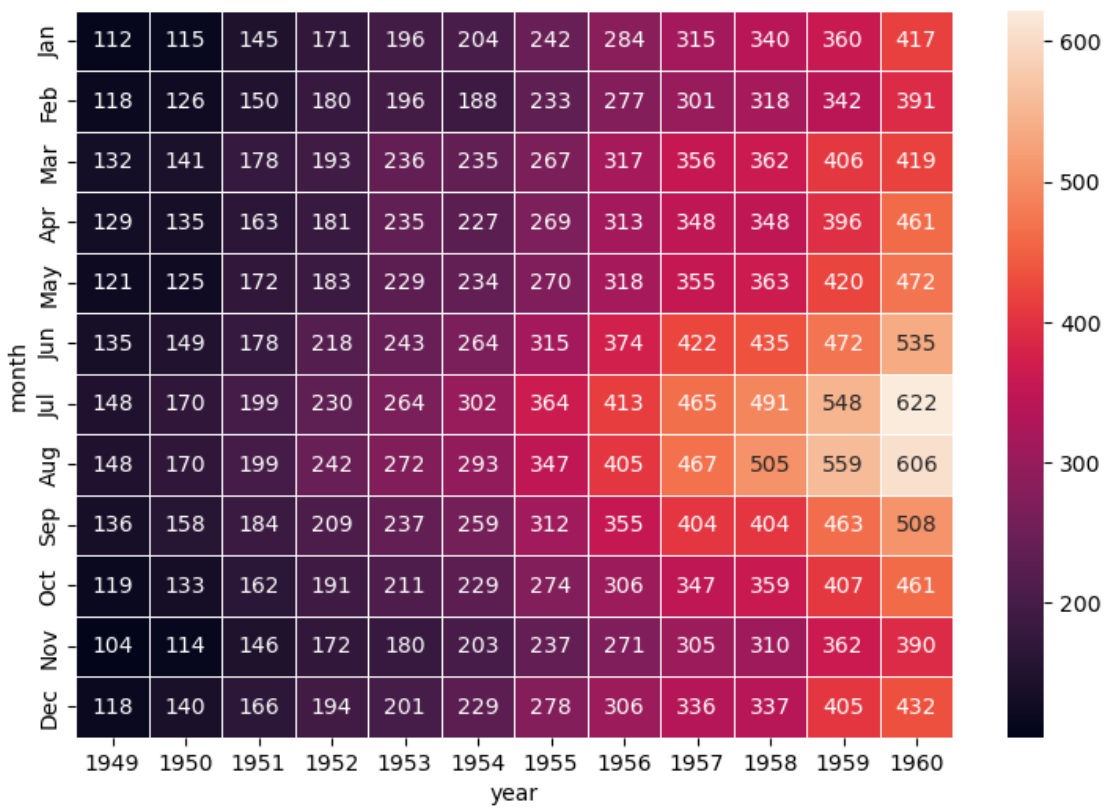
[31]:
```
# Load the example flights dataset and convert to long-form

# Load flights dataset
flights_long = sns.load_dataset("flights")

# Pivot to wide format
flights = flights_long.pivot(index="month", columns="year", values="passengers")

print(flights.head(5))
```

```
year    1949  1950  1951  1952  1953  1954  1955  1956  1957  1958  1959  1960
```

```
month
Jan      112   115   145   171   196   204   242   284   315   340   360   417
Feb      118   126   150   180   196   188   233   277   301   318   342   391
Mar      132   141   178   193   236   235   267   317   356   362   406   419
Apr      129   135   163   181   235   227   269   313   348   348   396   461
May      121   125   172   183   229   234   270   318   355   363   420   472
```

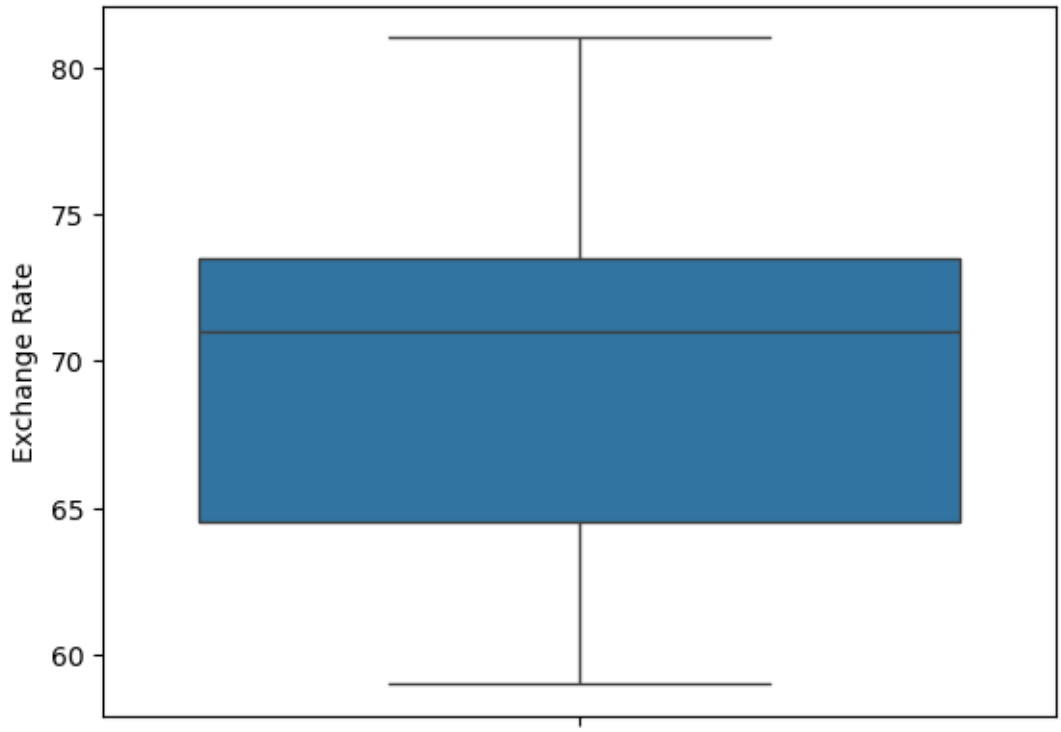
[32]:
```
# Draw a heatmap with the numeric values in each cell

f, ax = plt.subplots(figsize=(9, 6))
sns.heatmap(flights, annot=True, fmt='d', linewidths=.5, ax=ax)
```

[32]: <Axes: xlabel='year', ylabel='month'>

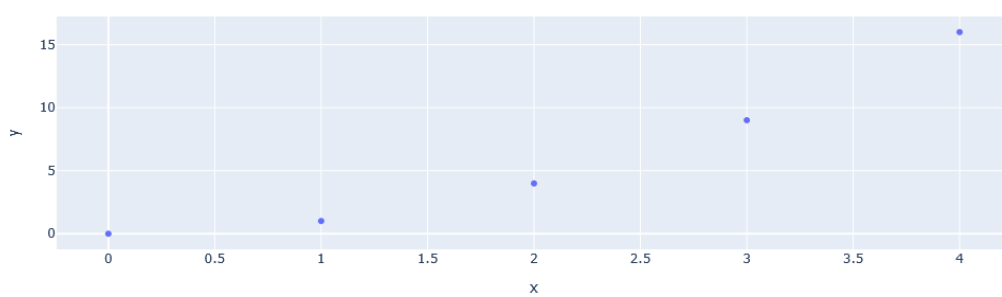


[33]:
```
sns.boxplot(y=df["Exchange Rate"])
```

[33]: <Axes: ylabel='Exchange Rate'>



[34]:
```
# x and y given as array_like objects

import plotly.express as px
fig = px.scatter(x = [0, 1, 2, 3, 4], y = [0, 1, 4, 9, 16])
fig.show()
```



[ ]:
```

```